

EDGE-DETECTION

Aim:

To perform edge detection using Sobel, Laplacian, and Canny edge detectors.

Software Required:

Anaconda - Python 3.7

Algorithm:

Step1:

Import all the necessary modules for the program.

Step2:

Load a image using imread() from cv2 module.

Step3:

Convert the image to grayscale

Step4:

Using Sobel operator from cv2,detect the edges of the image.

Step5:

Using Laplacian operator from cv2,detect the edges of the image and Using Canny operator from cv2,detect the edges of the image.

Program :

```
import cv2
import matplotlib.pyplot as plt

# 1. Read the image using imread
# Replace 'your_image.jpg' with your actual file path
img = cv2.imread('rose.png')

# 2. Convert the color (Note: OpenCV reads in BGR, so we convert BGR to RGB for Matplotlib)
gray = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
gray_single_channel = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY) # Often used for processing
gray = cv2.GaussianBlur(gray, (3,3), 0)
```

```
# --- Sobel X ---
sobelx = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=5)
plt.figure(figsize=(8,8))
plt.subplot(1,2,1)
plt.imshow(gray)
plt.title("Original Image")
plt.axis("off")
plt.subplot(1,2,2)
plt.imshow(sobelx)
plt.title("Sobel X axis")
plt.axis("off")
plt.show()

# --- Sobel Y ---
# 3. Complete the Sobel Y code
sobely = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=5)
plt.figure(figsize=(8,8))
plt.subplot(1,2,1)
plt.imshow(gray)
plt.title("Original Image")
plt.axis("off")
plt.subplot(1,2,2)
plt.imshow(sobely)
plt.title("Sobel Y axis")
plt.axis("off")
plt.show()

# --- Sobel XY ---
sobelxy = cv2.Sobel(gray, cv2.CV_64F, 1, 1, ksize=5)
plt.figure(figsize=(8,8))
plt.subplot(1,2,1)
plt.imshow(gray)
plt.title("Original Image")
plt.axis("off")
# 4. Add the subplot command for the second image
plt.subplot(1,2,2)
plt.imshow(sobelxy)
plt.title("Sobel XY axis")
plt.axis("off")
plt.show()

# --- Laplacian ---
lap = cv2.Laplacian(gray, cv2.CV_64F)
plt.figure(figsize=(8,8))
plt.subplot(1,2,1)
plt.imshow(gray)
plt.title("Original Image")
plt.axis("off")
plt.subplot(1,2,2)
# 5. Display the image using imshow
plt.imshow(lap)
plt.title("Laplacian Edge Detector")
plt.axis("off")
plt.show()

# --- Canny ---
```

```
canny = cv2.Canny(img, 120, 150) # Canny usually works best on the original or grayscale
plt.figure(figsize=(8,8))
plt.subplot(1,2,1)
plt.imshow(gray)
plt.title("Original Image")
plt.axis("off")
plt.subplot(1,2,2)
plt.imshow(canny, cmap='gray')
# 6. Provide the title
plt.title("Canny Edge Detection")
plt.axis("off")
plt.show()
```

Output:

SOBEL EDGE DETECTOR

Original Image



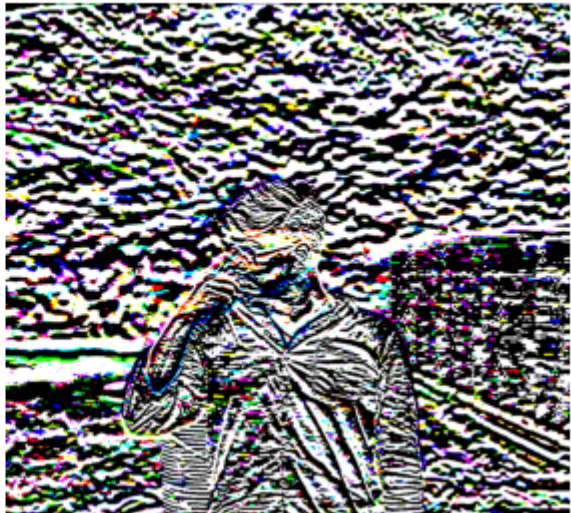
Sobel X axis



Original Image



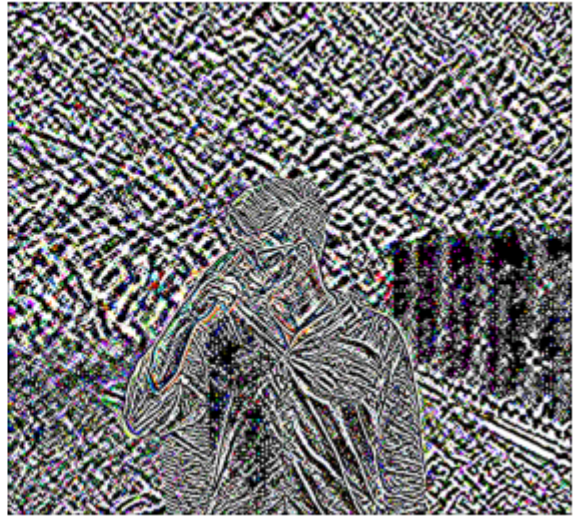
Sobel Y axis



Original Image



Sobel XY axis



LAPLACIAN EDGE DETECTOR

Original Image



Laplacian Edge Detector



CANNY EDGE DETECTOR

Original Image



Canny Edge Detection



Result: Thus the edges are detected using Sobel, Laplacian, and Canny edge detectors.